Combinatorics of Saddle Points in Deep Linear Networks

Abstract

We study the rank-r critical loci of deep linear networks directly in weight space. Fixing a singular subset S of the teacher matrix determines a closing map

$$W_0 = \Sigma_{XY} P_Q,$$

and the transpose-gradient equations define a cyclic locus $\Gamma_{d,S} \subseteq \Omega$. Its rank-$r$ critical locus $\mathcal{C}^r_{d,S}$ is the distinguished fiber cut out by the additional constraint

$$\mu(W) = P_S W^*.$$

Our first result is a direct weight-space orbit correspondence: $\mathcal{C}^r_{d,S}/H_S$ is in bijection with $\Gamma^r_{d,S}/G_d^{W_0}$.

We then identify $\Gamma_{d,S}$ with a fixed-arrow representation variety of the weight-space cyclic Nakayama algebra A_L^{wt} and classify its indecomposable modules. This yields an orbit stratification of the cyclic locus by multiplicity patterns. To obtain the combinatorics relevant for saddle points, we introduce lifted cyclic rank patterns on the universal cover of the cycle and prove that multiplicity patterns and rank patterns are equivalent via explicit forward and inversion formulas. As a consequence, the cyclic locus and the critical locus are both stratified by admissible cyclic rank patterns.

Finally, because every stratum is a single group orbit, we derive exact codimension formulas from orbit dimensions. The codimension of the critical locus is obtained by minimizing an explicit quadratic functional over the admissible cyclic rank patterns, and it differs from the codimension of the rank-r cyclic locus by the universal endpoint term

$$r(d_0 - d_L + r).$$

In the full-rank case $r = d_L$, our results recover the global-minimum fiber geometry studied by Pepin Lehalleur and Rimányi.

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1 Introduction

Critical points of deep linear networks are highly structured and highly degenerate. Unlike Morse-type landscapes, the relevant objects are not isolated points but algebraic loci carrying large symmetry groups. For the purpose of understanding saddle points, one therefore needs a discrete combinatorial label for an entire critical stratum rather than a local description of a single point.

Two recent developments motivate the present work. Achour, Malgouyres, and Gerchinovitz gave a complete second-order classification of critical points of deep linear networks and showed that the vanishing-gradient equations can be rewritten as cyclic vanishing relations [1]. On the other hand, Pepin Lehalleur and Rimányi studied the fiber of the multiplication map at a fixed end-to-end matrix and gave a quiver-theoretic description of the geometry of global minima [3]. The aim of the present paper is to connect these two viewpoints and obtain a combinatorial description of saddle strata of fixed rank.

The main structural choice in this draft is to work directly in the parameter space

$$\Omega = \prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R})$$

of the network. Previous versions of the project encoded the cyclic equations in a residual parameter space and then lifted the resulting data back to weight space. Here we keep both the cyclic locus and the critical locus inside Ω from the start and express their relation by an orbit correspondence. This direct weight-space formulation is better suited to saddle combinatorics, because the labels that classify the cyclic locus transfer immediately to the critical locus.

Contributions

The paper makes four contributions.

1. We prove a direct weight-space orbit correspondence between the cyclic locus and the critical locus:
$$\mathcal{C}^r_{d,S}/H_S \xrightarrow{\sim} \Gamma^r_{d,S}/G_d^{W_0}.$$

2. We identify the fixed-arrow cyclic locus with a representation variety of a cyclic monomial Nakayama algebra A_L^{wt}, classify its indecomposable modules, and recover the orbit stratification by multiplicity patterns.

3. We introduce lifted cyclic rank patterns on the universal cover of the cycle, prove an explicit bijection between multiplicity patterns and rank patterns, and deduce a stratification of both $\Gamma^r_{d,S}$ and $\mathcal{C}^r_{d,S}$ by admissible cyclic rank patterns.

4. We derive exact codimension formulas for the cyclic and critical strata by orbit-dimension arguments. The codimension of the critical locus is obtained by minimizing an explicit quadratic functional on the admissible rank-pattern set.

Related work

The starting point for the critical-point analysis is the work of Achour, Malgouyres, and Gerchinovitz [1], who classify critical points of deep linear networks at second order and show how the singular-value decomposition of the teacher matrix controls the critical values. Our use of the cyclic equations follows this line.

The global-minimum benchmark comes from Pepin Lehalleur and Rimányi [3], who study fibers of the multiplication map by quiver-representation methods. Their work provides the correct full-rank model for the present paper: when $r = d_L$, the closing map vanishes, the cyclic locus becomes the full-rank determinantal locus, and our saddle-stratification results specialize to the orbit geometry of the multiplication-map fiber.

On the representation-theoretic side, we use only standard facts on bound quivers, Nakayama algebras, and Krull–Schmidt decompositions; a general reference is [2].

Organization

Section 2 introduces the direct weight-space setup. Section 3 proves the orbit correspondence between the cyclic and critical loci. Section 4 identifies the cyclic locus with a fixed-arrow representation variety and classifies its orbits by multiplicity patterns. Section 5 replaces multiplicity labels by cyclic rank patterns. Section 6 transfers the rank-pattern stratification to the critical locus, giving the main theorem of the paper. Section 7 gives exact codimension formulas. The appendices collect a detailed inversion proof and the homological comparison with the older Ext^1-based viewpoint.

Whenever a proposition has been checked in Lean in the companion formalization, we state this explicitly after the proof.

2 Setup and Background

2.1 Notation

Notation	Meaning
L	network depth, i.e. number of weight matrices
$d = (d_0, \dots, d_L)$	dimension vector
Ω	parameter space $\prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R})$
μ	product map $\mu(W) = W_L \cdots W_1$
W^*	teacher matrix $\Sigma_{YX}\Sigma_X^{-1}$
$r_{\max}$	rank of W^*
$S \subseteq \{1, \dots, r_{\max}\}$	singular subset, with $\lvert S\rvert = r$
P_S, P_Q	complementary orthogonal projectors on $\mathbb{R}^{d_L}$
W_0	closing map $W_0 = \Sigma_{XY} P_Q$
Σ_d^r	rank-r determinantal locus in Ω
$\Gamma_{d,S}$	weight-space cyclic locus
$\Gamma_{d,S}^r$	rank-r cyclic locus
$\mathcal{C}_{d,S}^r$	rank-r critical locus attached to S
G_d	full base-change group $\prod_{\ell=0}^{L} \mathrm{GL}(d_\ell, \mathbb{R})$
$G_d^{W_0}$	stabilizer of W_0 in G_d
H_S	stabilizer in $G_d^{W_0}$ of $P_S W^*$
A_L^{wt}	weight-space cyclic Nakayama algebra
$\mathcal{I}_L^{\mathrm{wt}}$	lifted admissible interval index set
$\mathcal{M}_d^+(A_L^{\mathrm{wt}})$	multiplicity-pattern set
$\mathcal{R}_d^{\mathrm{wt}}$	lifted cyclic rank-pattern set

2.2 Deep linear networks

Definition 2.1 (Deep linear network and product map)**.** Let

$$\Omega := \prod_{\ell=1}^{L} \mathrm{Mat}_{d_\ell \times d_{\ell-1}}(\mathbb{R}), \qquad W = (W_1, \dots, W_L) \in \Omega.$$

The associated deep linear network is

$$x \longmapsto W_L \cdots W_1 x,$$

and its end-to-end matrix is given by the product map

$$\mu : \Omega \to \mathrm{Mat}_{d_L \times d_0}(\mathbb{R}), \qquad \mu(W) := W_L \cdots W_1.$$

For each $1 \le \ell \le L$, we use the partial-product notation

$$W_{>\ell} := W_L \cdots W_{\ell+1} \in \mathrm{Mat}_{d_L \times d_\ell}(\mathbb{R}), \qquad W_{<\ell} := W_{\ell-1} \cdots W_1 \in \mathrm{Mat}_{d_{\ell-1} \times d_0}(\mathbb{R}),$$

with the conventions

$$W_{>L} = I_{d_L}, \qquad W_{<1} = I_{d_0}.$$

Definition 2.2 (Population loss and teacher matrix)**.** Let

$$\mathcal{L}(W) := \mathbb{E}\big[\|Y - \mu(W)X\|_2^2\big].$$

Write

$$\Sigma_X = \mathbb{E}[XX^\top], \qquad \Sigma_{XY} = \mathbb{E}[XY^\top], \qquad \Sigma_{YX} = \Sigma_{XY}^\top.$$

The teacher matrix is

$$W^* := \Sigma_{YX}\Sigma_X^{-1}, \qquad r_{\max} := \operatorname{rank}(W^*).$$

Standing assumptions. Throughout the paper we assume that Σ_X is invertible, that Σ_{XY} has full rank d_L, that W^* has distinct nonzero singular values, and that the network has no bottleneck:

$$d_\ell \geq d_L \qquad \text{for every } 0 \leq \ell \leq L.$$

2.3 Gradient equations in weight space

Proposition 2.3 (Gradient formula)**.** For each $1 \leq \ell \leq L$,

$$\nabla_{W_\ell}\mathcal{L}(W) = 2\, W_{>\ell}^\top\big(\mu(W)\Sigma_X - \Sigma_{YX}\big)W_{<\ell}^\top.$$

Consequently,

$$\big(\nabla_{W_\ell}\mathcal{L}(W)\big)^\top = 2\, W_{<\ell}\big(\Sigma_X\mu(W)^\top - \Sigma_{XY}\big)W_{>\ell}.$$

Proof. The differential of the product map in the direction $\dot{W}_\ell$ is

$$D\mu(W)[\dot{W}_\ell] = W_{>\ell}\dot{W}_\ell W_{<\ell}.$$

The gradient of the quadratic loss with respect to the end-to-end matrix M is

$$\nabla_M\mathcal{L}(M) = 2(M\Sigma_X - \Sigma_{YX}),$$

so the chain rule gives the first identity. The second follows by taking the transpose. □

Fix a singular subset $S \subseteq \{1, \ldots, r_{\max}\}$ of cardinality $|S| = r$, and let

$$P_S,\ P_Q$$

be the corresponding complementary orthogonal projectors on $\mathbb{R}^{d_L}$. Define

$$W_0 := \Sigma_{XY}P_Q.$$

Proposition 2.4 (Weight-space cyclic equations)**.** Let W be a critical point of rank r attached to S, so that

$$\mu(W) = P_S W^*.$$

Then the critical-point equations are equivalent to

$$W_{<\ell}W_0W_{>\ell} = 0 \qquad \text{for every } 1 \leq \ell \leq L.$$

Proof. At a critical point attached to S,

$$\mu(W) = P_S W^* = P_S \Sigma_{YX} \Sigma_X^{-1}.$$

Hence

$$\Sigma_X \mu(W)^\top - \Sigma_{XY} = \Sigma_X \Sigma_X^{-1} \Sigma_{XY} P_S - \Sigma_{XY} = \Sigma_{XY}(P_S - I) = -\Sigma_{XY} P_Q = -W_0.$$

Substituting this identity into the transpose-gradient formula yields the claimed equations. □

Definition 2.5 (Determinantal, cyclic, and critical loci)**.** Define

$$\Sigma_d^r := \{W \in \Omega : \operatorname{rank}(\mu(W)) = r\},$$

$$\Gamma_{d,S} := \{W \in \Omega \mid W_{<\ell} W_0 W_{>\ell} = 0 \text{ for all } 1 \le \ell \le L\},$$

$$\Gamma_{d,S}^r := \Gamma_{d,S} \cap \Sigma_d^r,$$

and

$$\mathcal{C}_{d,S}^r := \left\{W \in \Gamma_{d,S}^r \;\middle|\; \mu(W) = P_S W^*\right\}.$$

3 From the Cyclic Locus to the Critical Locus

3.1 Symmetry groups

Definition 3.1 (Relevant symmetry groups)**.** The full base-change group is

$$G_d := \prod_{\ell=0}^{L} \mathrm{GL}(d_\ell, \mathbb{R}),$$

acting on Ω by

$$(g \cdot W)_\ell := g_\ell W_\ell g_{\ell-1}^{-1}.$$

The stabilizer of the closing map is

$$G_d^{W_0} := \{g = (g_0, \ldots, g_L) \in G_d \mid g_0 W_0 g_L^{-1} = W_0\}.$$

Finally, the stabilizer of the distinguished critical product is

$$H_S := \mathrm{Stab}_{G_d^{W_0}}(P_S W^*) = \{g \in G_d^{W_0} \mid g_L (P_S W^*) g_0^{-1} = P_S W^*\}.$$

Lemma 3.2 (Equivariance of the product map)**.** For every $g \in G_d$ and every $W \in \Omega$,

$$\mu(g \cdot W) = g_L \mu(W) g_0^{-1}.$$

Proof. Expand the product and cancel adjacent inverse factors:

$$\mu(g \cdot W) = (g_L W_L g_{L-1}^{-1}) \cdots (g_1 W_1 g_0^{-1}) = g_L W_L \cdots W_1 g_0^{-1} = g_L \mu(W) g_0^{-1}.$$

□

Proposition 3.3 (Stability of the cyclic locus)**.** The loci $\Gamma_{d,S}$ and $\Gamma_{d,S}^r$ are stable under $G_d^{W_0}$.

Proof. For fixed ℓ, set

$$E_\ell(W) := W_{<\ell} W_0 W_{>\ell}.$$

If $g \in G_d^{W_0}$, then

$$(g \cdot W)_{<\ell} = g_{\ell-1} W_{<\ell} g_0^{-1}, \qquad (g \cdot W)_{>\ell} = g_L W_{>\ell} g_\ell^{-1},$$

so

$$E_\ell(g \cdot W) = g_{\ell-1} W_{<\ell} (g_0^{-1} W_0 g_L) W_{>\ell} g_\ell^{-1} = g_{\ell-1} E_\ell(W) g_\ell^{-1}.$$

Thus the cyclic equations are preserved, and the rank of $\mu(W)$ is preserved by equivariance of μ. □

Proposition 3.4 (The critical locus is not $G_d^{W_0}$-stable for $r > 0$)**.** Assume $r > 0$. Then $\mathcal{C}_{d,S}^r$ is not stable under $G_d^{W_0}$.

Proof. Choose $\lambda \in \mathbb{R}^\times$ with $\lambda \neq 1$, and define

$$g_0 = I_{d_0}, \qquad g_L = \lambda P_S + P_Q, \qquad g_\ell = I_{d_\ell} \quad (1 \leq \ell \leq L-1).$$

Then

$$g_0 W_0 g_L^{-1} = \Sigma_{XY} P_Q (\lambda^{-1} P_S + P_Q) = \Sigma_{XY} P_Q = W_0,$$

so $g \in G_d^{W_0}$. If $W \in \mathcal{C}_{d,S}^r$, then

$$\mu(g \cdot W) = g_L(P_S W^*) = \lambda P_S W^* \neq P_S W^*.$$

Hence $g \cdot W \notin \mathcal{C}_{d,S}^r$. □

3.2 Endpoint reduction and normalization

Definition 3.5 (Endpoint group)**.** Define

$$K_S := \{(g_0, g_L) \in \mathrm{GL}(d_0, \mathbb{R}) \times \mathrm{GL}(d_L, \mathbb{R}) \mid g_0 W_0 g_L^{-1} = W_0\},$$

acting on $\mathrm{Mat}_{d_L \times d_0}(\mathbb{R})$ by

$$(g_0, g_L) \cdot M := g_L M g_0^{-1}.$$

Lemma 3.6 (Endpoint projection)**.** The projection

$$\pi : G_d^{W_0} \to K_S, \qquad \pi(g_0, \ldots, g_L) := (g_0, g_L),$$

is surjective. Moreover, for every $W \in \Omega$,

$$\mu(G_d^{W_0} \cdot W) = K_S \cdot \mu(W).$$

Proof. If $(g_0, g_L) \in K_S$, then

$$(g_0, I_{d_1}, \ldots, I_{d_{L-1}}, g_L) \in G_d^{W_0},$$

so π is surjective. The orbit identity follows from equivariance of μ. □

Corollary 3.7 (Endpoint reduction)**.** For $W \in \Gamma_{d,S}^r$, the following are equivalent:

1. there exists $g \in G_d^{W_0}$ such that $g \cdot W \in \mathcal{C}_{d,S}^r$;

2. there exists $(g_0, g_L) \in K_S$ such that

$$g_L \mu(W) g_0^{-1} = P_S W^*;$$

3. $\mu(W) \in K_S \cdot (P_S W^*)$.

Proof. This is immediate from the endpoint projection lemma and the definition of the critical locus. □

Proposition 3.8 (Visible product-level constraints)**.** If $W \in \Gamma_{d,S}$ and $M = \mu(W)$, then

$$W_0 M = 0 \qquad \text{and} \qquad M W_0 = 0.$$

Consequently,

$$\mathrm{im}(M) \subseteq \mathrm{Ker}(W_0), \qquad \mathrm{im}(W_0) \subseteq \mathrm{Ker}(M).$$

Proof. For $\ell = 1$, the cyclic equation reads

$$W_0 W_{>1} = 0.$$

Multiplying on the right by W_1 gives $W_0 M = 0$. For $\ell = L$, the cyclic equation reads $W_{<L} W_0 = 0$. Multiplying on the left by W_L gives $M W_0 = 0$. □

Corollary 3.9 (Rank-r sharpening)**.** Assume Σ_{XY} has full rank d_L. If $W \in \Gamma^r_{d,S}$ and $M = \mu(W)$, then

$$\mathrm{Ker}(W_0) = \mathrm{im}(P_S), \qquad \mathrm{rank}(W_0) = d_L - r, \qquad \mathrm{im}(M) = \mathrm{im}(P_S).$$

Proof. Because Σ_{XY} has full rank d_L, it is injective as a map from $\mathbb{R}^{d_L}$ to $\mathbb{R}^{d_0}$. Hence

$$\mathrm{Ker}(W_0) = \mathrm{Ker}(\Sigma_{XY} P_Q) = \mathrm{Ker}(P_Q) = \mathrm{im}(P_S).$$

Since M has rank r and $\mathrm{im}(M) \subseteq \mathrm{Ker}(W_0)$, the inclusion is an equality. □

Theorem 3.10 (Normalization theorem)**.** Assume that Σ_{XY} has full rank d_L. Then for every

$$W \in \Gamma^r_{d,S}$$

one has

$$\mu(W) \in K_S \cdot (P_S W^*).$$

Proof. Set

$$\sigma := W_0, \qquad T := P_S W^*, \qquad U := \mathrm{Ker}(\sigma) \subseteq \mathbb{R}^{d_L}, \qquad I := \mathrm{im}(\sigma) \subseteq \mathbb{R}^{d_0}.$$

By the previous proposition and corollary, if $M = \mu(W)$ with $W \in \Gamma^r_{d,S}$, then

$$\sigma M = 0, \qquad M\sigma = 0, \qquad \mathrm{im}(M) = U.$$

Therefore M induces a surjective map

$$\overline{M} : \mathbb{R}^{d_0}/I \to U.$$

The distinguished matrix $T = P_S W^*$ satisfies the same two annihilation relations and has image U. Indeed, $W_0 T = 0$ follows from $P_Q P_S = 0$, and $TW_0 = 0$ is part of the rank-r critical-point description of [1]; hence T induces a surjective map

$$\overline{T} : \mathbb{R}^{d_0}/I \to U.$$

Now every element of K_S preserves U and I. Conversely, by choosing complements of U and I, one checks that the induced action of K_S on reduced maps is the natural left-right action of

$$\mathrm{GL}(U) \times \mathrm{GL}(\mathbb{R}^{d_0}/I),$$

and that this action is surjective. Since $\overline{M}$ and $\overline{T}$ are both surjective maps from $\mathbb{R}^{d_0}/I$ onto U, they lie in the same $\mathrm{GL}(U) \times \mathrm{GL}(\mathbb{R}^{d_0}/I)$-orbit. Lifting back to K_S yields

$$M \in K_S \cdot T.$$

□

3.3 Orbit correspondence

Theorem 3.11 (Orbit-surjectivity)**.** Every $G_d^{W_0}$-orbit in $\Gamma_{d,S}^r$ meets the critical locus $\mathcal{C}_{d,S}^r$.

Proof. Let $W \in \Gamma_{d,S}^r$. By Theorem 3.10,

$$\mu(W) \in K_S \cdot (P_S W^*).$$

Endpoint reduction therefore gives $g \in G_d^{W_0}$ such that

$$g \cdot W \in \mathcal{C}_{d,S}^r.$$

□

Theorem 3.12 (Direct weight-space orbit correspondence)**.** Every $G_d^{W_0}$-orbit in $\Gamma_{d,S}^r$ meets $\mathcal{C}_{d,S}^r$ in exactly one H_S-orbit. Equivalently, the inclusion

$$\mathcal{C}_{d,S}^r \hookrightarrow \Gamma_{d,S}^r$$

induces a bijection

$$\mathcal{C}_{d,S}^r / H_S \xrightarrow{\sim} \Gamma_{d,S}^r / G_d^{W_0}.$$

Proof. Existence is orbit-surjectivity. For uniqueness, let $W, W' \in \mathcal{C}_{d,S}^r$ lie in the same $G_d^{W_0}$-orbit, say $W' = g \cdot W$. Since both points lie in the critical locus,

$$\mu(W) = P_S W^* = \mu(W').$$

By equivariance of μ,

$$P_S W^* = g_L (P_S W^*) g_0^{-1},$$

so $g \in H_S$. Thus

$$\left(G_d^{W_0} \cdot W\right) \cap \mathcal{C}_{d,S}^r = H_S \cdot W.$$

□

Corollary 3.13 (Transfer of stratifications)**.** Any decomposition of $\Gamma_{d,S}^r$ into locally closed $G_d^{W_0}$-stable unions of orbits transfers, by intersection with $\mathcal{C}_{d,S}^r$, to a decomposition of $\mathcal{C}_{d,S}^r$ into locally closed H_S-stable unions of orbits.

4 The Weight-Space Cyclic Algebra and Multiplicity Patterns

4.1 Bound-quiver description

Definition 4.1 (The cyclic quiver and the weight-space ideal)**.** Let

$$Q = C_{L+1}$$

be the oriented cycle on vertices

$$Q_0 = \{0, 1, \dots, L\}$$

with arrows

$$a_t : t-1 \to t \qquad (1 \le t \le L), \qquad a_0 : L \to 0.$$

For each $\ell \in \{1, \dots, L\}$, let

$$p_\ell := a_{\ell-1} \cdots a_1 a_0 a_L \cdots a_{\ell+1},$$

the unique length-L path omitting the arrow a_ℓ. Define

$$I_L^{\mathrm{wt}} := \langle p_1, \dots, p_L \rangle \subseteq kQ, \qquad A_L^{\mathrm{wt}} := kQ/I_L^{\mathrm{wt}}.$$

Remark 4.2. The only length-L path not killed by I_L^{wt} is

$$p_0 := a_L \cdots a_1.$$

In the DLN dictionary this is exactly the open-chain product $\mu(W)$, so it must survive in the direct weight-space formalism.

Definition 4.3 (Representation varieties)**.** Fix vector spaces $V_i \cong k^{d_i}$. Set

$$\mathrm{Rep}_d(Q) := \prod_{\ell=1}^{L} \mathrm{Hom}(V_{\ell-1}, V_\ell) \times \mathrm{Hom}(V_L, V_0),$$

and

$$\mathrm{Rep}_d(A_L^{\mathrm{wt}}) := \{\phi \in \mathrm{Rep}_d(Q) \mid p_\ell(\phi) = 0 \text{ for all } 1 \le \ell \le L\}.$$

For a fixed map $W_0 \in \mathrm{Hom}(V_L, V_0)$, define the fixed-arrow slice

$$\mathrm{Rep}_{d,W_0}(Q, I_L^{\mathrm{wt}}) := \{\phi \in \mathrm{Rep}_d(A_L^{\mathrm{wt}}) \mid \phi_0 = W_0\}.$$

Proposition 4.4 (The cyclic locus is a fixed-arrow representation variety)**.** Specialize to $k = \mathbb{R}$ and fix

$$W_0 = \Sigma_{XY} P_Q.$$

For $W = (W_1, \dots, W_L) \in \Omega$, define

$$\phi(W) := (W_0, W_1, \dots, W_L) \in \mathrm{Rep}_d(Q).$$

Then

$$W \in \Gamma_{d,S} \iff \phi(W) \in \mathrm{Rep}_{d,W_0}(Q, I_L^{\mathrm{wt}}).$$

Proof. For each $1 \le \ell \le L$,

$$p_\ell(\phi(W)) = W_{\ell-1} \cdots W_1 W_0 W_L \cdots W_{\ell+1} = W_{<\ell} W_0 W_{>\ell}.$$

Thus the quiver relations are exactly the weight-space cyclic equations. □

4.2 The weight-space algebra

Lemma 4.5 (Surviving paths)**.** In A_L^{wt}, the nonzero oriented paths are precisely the path classes represented by

$$(a,b) \in \mathcal{I}_L^{\mathrm{wt}} := \{(a,b) \in \mathbb{Z}^2 \mid 0 \le a \le L,\ a \le b \le a+L\} \setminus \{(a,a+L) \mid 1 \le a \le L\}.$$

Proof. Every oriented path on the cycle lifts uniquely to an interval $[a,b]$ on the universal cover with $0 \le a \le L$ and $a \le b \le a+L$. The relation ideal kills exactly the length-L paths $(a,a+L)$ with $1 \le a \le L$, and all longer paths contain such a subpath. The unique surviving length-L path is $(0,L)$. □

Corollary 4.6 (Projective indecomposables)**.** The indecomposable projectives are

$$P(0) = M(0,L), \qquad P(a) = M(a,a+L-1) \quad (1 \le a \le L).$$

Equivalently, the Kupisch series of A_L^{wt} is

$$(L+1, L, \dots, L).$$

Proof. By Lemma 4.5, the maximal nonzero path starting at 0 has length L, while the maximal nonzero path starting at $a > 0$ has length $L-1$. □

Theorem 4.7 (The weight-space algebra is Nakayama)**.** The algebra A_L^{wt} is a finite-dimensional cyclic monomial Nakayama algebra.

Proof. The ideal I_L^{wt} is monomial, every path of length $L+1$ is zero by Lemma 4.5, and the quiver is an oriented cycle. Hence the quotient is finite-dimensional, monomial, and Nakayama [2, Chapter VI]. □

Definition 4.8 (Interval modules)**.** For $(a,b) \in \mathcal{I}_L^{\mathrm{wt}}$, let $M(a,b)$ be the interval representation supported on the lifted interval $[a,b]$: it is one-dimensional on the vertices visited by $[a,b]$, every arrow inside the support acts by the identity, and every arrow outside the support acts by zero.

Theorem 4.9 (Classification of indecomposables)**.** The finite-dimensional indecomposable A_L^{wt}-modules are exactly the interval modules

$$M(a,b) \qquad ((a,b) \in \mathcal{I}_L^{\mathrm{wt}}),$$

and each indecomposable occurs uniquely in this form.

Proof. By Theorem 4.7, every indecomposable is a quotient of a unique indecomposable projective. Using Corollary 4.6, these quotients are precisely the interval modules $M(a,b)$ indexed by the admissible lifted intervals. □

4.3 Multiplicity patterns and orbit classification

Definition 4.10 (Multiplicity patterns)**.** Define

$$\mathcal{M}_d^+(A_L^{\mathrm{wt}})$$

to be the set of arrays

$$m = (m_{ab})_{(a,b)\in\mathcal{I}_L^{\mathrm{wt}}} \in \mathbb{Z}_{\ge 0}^{\mathcal{I}_L^{\mathrm{wt}}}$$

such that the module

$$\bigoplus_{(a,b)\in\mathcal{I}_L^{\mathrm{wt}}} M(a,b)^{\oplus m_{ab}}$$

has dimension vector d.

Proposition 4.11 (Krull–Schmidt decomposition)**.** Every finite-dimensional A_L^{wt}-module of dimension vector d decomposes uniquely as

$$M \cong \bigoplus_{(a,b)\in\mathcal{I}_L^{\mathrm{wt}}} M(a,b)^{\oplus m_{ab}}$$

for a unique multiplicity pattern

$$m = (m_{ab}) \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}).$$

Proof. The algebra is finite-dimensional, hence artinian, so its module category is Krull–Schmidt. The indecomposable classification of Theorem 4.9 yields the stated decomposition. □

Proposition 4.12 (Ambient and fixed-arrow orbit classification)**.** Two points of $\mathrm{Rep}_d(A_L^{\mathrm{wt}})$ lie in the same G_d-orbit if and only if they have the same multiplicity pattern. Two points of $\mathrm{Rep}_{d,W_0}(Q, I_L^{\mathrm{wt}})$ lie in the same $G_d^{W_0}$-orbit if and only if they have the same multiplicity pattern.

Proof. In both cases, belonging to the same orbit is equivalent to being isomorphic as A_L^{wt}-modules. By Proposition 4.11, this is equivalent to equality of multiplicity patterns. □

Definition 4.13 (The module attached to a cyclic point)**.** For $W \in \Gamma_{d,S}$, let

$$M(W)$$

be the A_L^{wt}-module corresponding to $\phi(W)$ under Proposition 4.4, and let

$$m_W = (m_{ab}(W)) \in \mathcal{M}_d^+(A_L^{\mathrm{wt}})$$

be its multiplicity pattern.

Definition 4.14 (Closing-arrow rank functional)**.** For $m \in \mathcal{M}_d^+(A_L^{\mathrm{wt}})$, define

$$\nu_0(m) := \sum_{\substack{(a,b)\in\mathcal{I}_L^{\mathrm{wt}} \\ a\le L<L+1\le b}} m_{ab}.$$

Equivalently, $\nu_0(m)$ is the total multiplicity of the indecomposable intervals whose support contains the closing arrow.

Proposition 4.15 (Realized multiplicity patterns)**.** The realized multiplicity patterns in the fixed-arrow cyclic locus are exactly those satisfying

$$\nu_0(m) = \mathrm{rank}(W_0) = d_L - r.$$

Proof. On an indecomposable interval $M(a,b)$, the closing arrow a_0 acts with rank 1 if and only if the interval contains the lifted segment $[L, L+1]$, and otherwise it acts with rank 0. Therefore the rank of the closing-arrow map on the direct sum defined by m is exactly $\nu_0(m)$.

If $m = m_W$ is realized by a point $W \in \Gamma_{d,S}$, then the closing arrow is exactly W_0, so necessarily $\nu_0(m) = \mathrm{rank}(W_0)$.

Conversely, if $\nu_0(m) = \mathrm{rank}(W_0)$, then the closing-arrow map of the module defined by m has the same rank as W_0. Since two real matrices are equivalent under $\mathrm{GL}(d_0) \times \mathrm{GL}(d_L)$ if and only if they have the same rank, an endpoint base change carries the closing arrow to W_0. Transporting the remaining arrows by the same base change yields a representative in the fixed-arrow slice. □

Theorem 4.16 (Multiplicity-pattern stratification of the cyclic locus)**.** The assignment

$$W \longmapsto m_W$$

induces a bijection

$$G_d^{W_0} \backslash \Gamma_{d,S} \xrightarrow{\sim} \{m \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}) \mid \nu_0(m) = d_L - r\}.$$

Equivalently,

$$\Gamma_{d,S} = \bigsqcup_{\substack{m \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}) \\ \nu_0(m) = d_L - r}} \Gamma_{d,S}[m], \qquad \Gamma_{d,S}[m] := \{W \in \Gamma_{d,S} \mid m_W = m\},$$

and each nonempty stratum $\Gamma_{d,S}[m]$ is a single $G_d^{W_0}$-orbit.

Proof. Combine Proposition 4.12 with Proposition 4.15. □

5 Rank Patterns and Cyclic Rank Stratification

5.1 Lifted cyclic rank patterns

Definition 5.1 (Lifted path coordinates)**.** For $(a, b) \in \mathcal{I}_L^{\mathrm{wt}}$, let

$$q_{ab}$$

denote the unique nonzero oriented path of A_L^{wt} represented by the lifted interval $[a, b]$.

Definition 5.2 (Lifted cyclic rank pattern)**.** Let M be an A_L^{wt}-module of dimension vector d. Its lifted cyclic rank pattern is the array

$$\rho(M) = (\rho_{ab}(M))_{(a,b) \in \mathcal{I}_L^{\mathrm{wt}}}, \qquad \rho_{ab}(M) := \mathrm{rank}\big(q_{ab}(M)\big).$$

Two coordinates play a distinguished role:

$$\rho_{0L}(M) = \mathrm{rank}(q_{0L}(M)), \qquad \rho_{L,L+1}(M) = \mathrm{rank}(q_{L,L+1}(M)).$$

In the DLN dictionary these are exactly the end-to-end rank and the rank of the closing arrow.

Proposition 5.3 (Forward formula)**.** Let M have multiplicity pattern

$$m = (m_{uv})_{(u,v) \in \mathcal{I}_L^{\mathrm{wt}}}.$$

Then for every $(a, b) \in \mathcal{I}_L^{\mathrm{wt}}$,

$$\rho_{ab}(M) = \sum_{\substack{(u,v) \in \mathcal{I}_L^{\mathrm{wt}} \\ u \le a \le b \le v}} m_{uv}.$$

Proof. On an indecomposable interval module $M(u, v)$, the path q_{ab} acts with rank 1 exactly when the path interval $[a, b]$ is contained in $[u, v]$, and otherwise with rank 0. The claim follows by additivity of rank on direct sums. □

Proposition 5.4 (Inversion formula)**.** Extend every rank pattern by the convention

$$\rho_{uv} = 0 \qquad \text{for } (u,v) \notin \mathcal{I}_L^{\mathrm{wt}}.$$

Then for every admissible interval $(a,b) \in \mathcal{I}_L^{\mathrm{wt}}$,

$$m_{ab} = \rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1}.$$

Proof. Insert the forward formula into the right-hand side and note that, for every fixed interval (u,v), the coefficient of m_{uv} is

$$\mathbf{1}_{u\le a\le b\le v} - \mathbf{1}_{u\le a-1\le b\le v} - \mathbf{1}_{u\le a\le b+1\le v} + \mathbf{1}_{u\le a-1\le b+1\le v}.$$

This coefficient equals 1 precisely when $(u,v) = (a,b)$ and equals 0 otherwise. A more detailed telescoping proof is recorded in Appendix A. □

Definition 5.5 (Admissible rank-pattern set)**.**

$$\mathcal{R}_d^{\mathrm{wt}} := \left\{ \rho \in \mathbb{Z}_{\ge 0}^{\mathcal{I}_L^{\mathrm{wt}}} \;\middle|\; \begin{array}{l} \rho_{aa} = d_a \text{ for every } 0 \le a \le L, \\ \rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1} \ge 0 \\ \qquad\qquad \text{for every } (a,b) \in \mathcal{I}_L^{\mathrm{wt}} \end{array} \right\}.$$

Theorem 5.6 (Multiplicity-rank bijection)**.** The multiplicity-pattern set and the rank-pattern set are canonically in bijection:

$$\mathcal{M}_d^+(A_L^{\mathrm{wt}}) \xrightarrow{\sim} \mathcal{R}_d^{\mathrm{wt}}.$$

Proof. The forward map sends a multiplicity pattern to the rank pattern given by Proposition 5.3. The inverse map is the finite-difference formula of Proposition 5.4. The diagonal condition $\rho_{aa} = d_a$ ensures that the resulting multiplicity pattern has dimension vector d. □

5.2 Fixed-arrow compatibility in rank coordinates

Proposition 5.7 (Fixed-arrow and rank-r constraints)**.** Let $W \in \Gamma_{d,S}$, and let $\rho(W)$ be its lifted cyclic rank pattern. Then

$$\rho_{L,L+1}(W) = \mathrm{rank}(W_0) = d_L - r.$$

If moreover $W \in \Gamma_{d,S}^r$, then

$$\rho_{0L}(W) = \mathrm{rank}(\mu(W)) = r.$$

Proof. The first identity is immediate from

$$q_{L,L+1}(W) = W_0,$$

and the second from

$$q_{0L}(W) = W_L \cdots W_1 = \mu(W).$$

□

Definition 5.8 (Admissible rank-r label set)**.** Define

$$\mathcal{R}_{d,S}^r := \{\rho \in \mathcal{R}_d^{\mathrm{wt}} \mid \rho_{L,L+1} = d_L - r,\ \rho_{0L} = r\}.$$

Theorem 5.9 (Rank-pattern stratification of the cyclic locus)**.** The assignment

$$W \longmapsto \rho(W)$$

induces bijections

$$G_d^{W_0} \backslash \Gamma_{d,S} \xrightarrow{\sim} \{\rho \in \mathcal{R}_d^{\mathrm{wt}} \mid \rho_{L,L+1} = d_L - r\},$$

and

$$G_d^{W_0} \backslash \Gamma_{d,S}^r \xrightarrow{\sim} \mathcal{R}_{d,S}^r.$$

Equivalently,

$$\Gamma_{d,S}^r = \bigsqcup_{\rho \in \mathcal{R}_{d,S}^r} \Gamma_{d,S}^r[\rho], \qquad \Gamma_{d,S}^r[\rho] := \{W \in \Gamma_{d,S}^r \mid \rho(W) = \rho\},$$

and every nonempty stratum is a single $G_d^{W_0}$-orbit.

Proof. Compose the multiplicity-pattern stratification theorem (Theorem 4.16) with the bijection of Theorem 5.6 and the scalar constraints of Proposition 5.7. □

Example 5.10 (Depth $L = 3$)**.** For $L = 3$, the lifted admissible intervals are

$$(0,0), (0,1), (0,2), (0,3), (1,1), (1,2), (1,3), (2,2), (2,3), (2,4), (3,3), (3,4), (3,5).$$

The wrapped intervals are $(2,4)$, $(3,4)$, and $(3,5)$, corresponding to the indecomposables containing the closing arrow. The two distinguished scalar coordinates are

$$\rho_{03} = \mathrm{rank}(W_3 W_2 W_1), \qquad \rho_{34} = \mathrm{rank}(W_0).$$

If $r = d_3$, then $\rho_{34} = 0$, so the wrapped intervals disappear and the admissible rank patterns reduce to the usual chain rank patterns for

$$0 \to 1 \to 2 \to 3.$$

6 Stratification of the Critical Loci

The orbit correspondence of Section 3 now transfers the cyclic rank-pattern labels to the critical locus.

Definition 6.1 (Critical rank-pattern strata)**.** For $\rho \in \mathcal{R}_{d,S}^r$, define

$$\mathcal{C}_{d,S}^r[\rho] := \mathcal{C}_{d,S}^r \cap \Gamma_{d,S}^r[\rho].$$

Theorem 6.2 (Main theorem: critical loci are stratified by cyclic rank patterns)**.** The rank-r critical locus admits the disjoint decomposition

$$\mathcal{C}_{d,S}^r = \bigsqcup_{\rho \in \mathcal{R}_{d,S}^r} \mathcal{C}_{d,S}^r[\rho].$$

Each nonempty stratum $\mathcal{C}_{d,S}^r[\rho]$ is a single H_S-orbit. Equivalently, the assignment

$$W \longmapsto \rho(W)$$

induces a bijection

$$H_S \backslash \mathcal{C}_{d,S}^r \xrightarrow{\sim} \mathcal{R}_{d,S}^r.$$

Proof. By Theorem 5.9, every nonempty stratum $\Gamma^r_{d,S}[\rho]$ is a single $G_d^{W_0}$-orbit. By the orbit correspondence theorem (Theorem 3.12), each such orbit meets $\mathcal{C}^r_{d,S}$ in exactly one H_S-orbit. Intersecting the cyclic decomposition with the critical locus yields the stated partition. □

Corollary 6.3 (Full-rank recovery)**.** If $r = d_L$, then

$$W_0 = 0, \qquad G_d^{W_0} = G_d, \qquad \Gamma^{d_L}_{d,S} = \Sigma_d^{d_L}.$$

In this case the admissible cyclic rank patterns are exactly the chain rank patterns of the equioriented quiver

$$0 \to 1 \to \cdots \to L,$$

and Theorem 6.2 recovers the orbit classification of the global-minimum fiber.

Proof. If $r = d_L$, then $P_Q = 0$, so the closing-arrow rank constraint is

$$\rho_{L,L+1} = 0.$$

Hence no wrapped interval occurs, and only the chain intervals remain. The orbit correspondence then becomes

$$\mathcal{C}^{d_L}_{d,S}/H_S \xrightarrow{\sim} \Sigma_d^{d_L}/G_d.$$

□

Remark 6.4 (Combinatorics of saddle strata)**.** Theorem 6.2 gives a discrete identity to each saddle stratum. Deep linear networks do not have isolated generic saddle points; instead, each saddle class is an H_S-orbit inside a positive-dimensional critical locus. The cyclic rank pattern is the finite combinatorial label that distinguishes these orbit classes.

7 Codimension Formulas

7.1 Hom spaces and the endomorphism functional

Proposition 7.1 (Hom criterion for interval modules)**.** For admissible intervals $(a, b), (c, d) \in \mathcal{I}_L^{\mathrm{wt}}$,

$$\dim_{\mathbb{R}} \mathrm{Hom}_A(M(a, b), M(c, d)) = \begin{cases} 1, & [a, b] \subseteq [c, d], \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Every interval module is one-dimensional on its support and all internal arrow maps are identities. A morphism is therefore determined by a single scalar on the source support and exists exactly when the source support is contained in the target support. □

Definition 7.2 (Endomorphism functional)**.** For

$$m = (m_{ab}) \in \mathcal{M}_d^+(A_L^{\mathrm{wt}}),$$

define

$$\mathsf{E}(m) := \sum_{(a,b)\in\mathcal{I}_L^{\mathrm{wt}}} \sum_{(c,d)\in\mathcal{I}_L^{\mathrm{wt}}} m_{ab} m_{cd}\, \mathbf{1}_{[a,b]\subseteq[c,d]}.$$

Proposition 7.3 (Endomorphism dimension)**.** If a module M has multiplicity pattern m, then

$$\dim_{\mathbb{R}} \mathrm{End}_A(M) = \mathsf{E}(m).$$

Proof. By additivity of Hom on finite direct sums and Proposition 7.1,

$$\dim \mathrm{End}_A(M) = \sum_{(a,b),(c,d)} m_{ab} m_{cd} \dim \mathrm{Hom}_A(M(a,b), M(c,d)),$$

which is exactly $\mathsf{E}(m)$. □

7.2 Dimensions of the acting groups

Proposition 7.4 (Dimension of $G_d^{W_0}$)**.** Let

$$s := \mathrm{rank}(W_0) = d_L - r.$$

Then

$$\dim G_d^{W_0} = \sum_{\ell=0}^{L} d_\ell^2 - s(d_0 + d_L - s) = \sum_{\ell=0}^{L} d_\ell^2 - (d_L - r)(d_0 + r).$$

Proof. The hidden-layer factors contribute $\sum_{\ell=1}^{L-1} d_\ell^2$. The endpoint factor is the stabilizer of a rank-s matrix under the left-right action of $\mathrm{GL}(d_0) \times \mathrm{GL}(d_L)$, hence has dimension

$$d_0^2 + d_L^2 - s(d_0 + d_L - s).$$

Adding the contributions gives the result. □

Proposition 7.5 (Dimension of H_S)**.** One has

$$\dim H_S = \sum_{\ell=0}^{L} d_\ell^2 - d_0 d_L.$$

Proof. Set

$$U := \mathrm{Ker}(W_0), \qquad I := \mathrm{im}(W_0).$$

Then

$$\dim U = r, \qquad \dim(\mathbb{R}^{d_0}/I) = d_0 - d_L + r.$$

By the normalization theorem, the quotient K_S/H_S is the left-right orbit of the reduced distinguished product

$$\overline{T} : \mathbb{R}^{d_0}/I \to U,$$

which is surjective of rank r. The orbit of a rank-r map between spaces of dimensions $d_0 - d_L + r$ and r has dimension

$$r(d_0 - d_L + r).$$

Thus

$$\dim H_S = \dim G_d^{W_0} - r(d_0 - d_L + r).$$

Substituting Proposition 7.4 and simplifying yields

$$\dim H_S = \sum_{\ell=0}^{L} d_\ell^2 - d_0 d_L.$$

□

7.3 Stratum codimension formulas

Definition 7.6 (Ambient dimension)**.** We write

$$\dim \Omega = \sum_{\ell=1}^{L} d_\ell d_{\ell-1}.$$

Theorem 7.7 (Codimension of the cyclic and critical strata)**.** Let $m \in \mathcal{M}_d^+(A_L^{\mathrm{wt}})$ be a multiplicity pattern realized in $\Gamma_{d,S}^r$. Then

$$\operatorname{codim}_\Omega \Gamma_{d,S}^r[m] = \Big(\sum_{\ell=1}^{L} d_\ell d_{\ell-1} - \sum_{\ell=0}^{L} d_\ell^2 + (d_L - r)(d_0 + r)\Big) + \mathsf{E}(m),$$

and

$$\operatorname{codim}_\Omega \mathcal{C}_{d,S}^r[m] = \Big(\sum_{\ell=1}^{L} d_\ell d_{\ell-1} - \sum_{\ell=0}^{L} d_\ell^2 + d_0 d_L\Big) + \mathsf{E}(m).$$

Proof. Let $W \in \Gamma_{d,S}^r[m]$. The stabilizer of W in $G_d^{W_0}$ is $\operatorname{Aut}_A(M(W))$, the module automorphism group of the corresponding A_L^{wt}-module. This is an open subset of the affine space $\operatorname{End}_A(M(W))$, hence has the same dimension:

$$\dim \operatorname{Aut}_A(M(W)) = \dim \operatorname{End}_A(M(W)) = \mathsf{E}(m)$$

by Proposition 7.3. Therefore

$$\dim \Gamma_{d,S}^r[m] = \dim(G_d^{W_0} \cdot W) = \dim G_d^{W_0} - \mathsf{E}(m).$$

Subtracting from $\dim \Omega$ and using Proposition 7.4 gives the cyclic formula.

Now let $W_c \in \mathcal{C}_{d,S}^r[m]$. By Theorem 6.2, $\mathcal{C}_{d,S}^r[m]$ is a single H_S-orbit. Its stabilizer in H_S is again $\operatorname{Aut}_A(M(W_c))$, so

$$\dim \mathcal{C}_{d,S}^r[m] = \dim H_S - \mathsf{E}(m).$$

Subtracting from $\dim \Omega$ and using Proposition 7.5 gives the critical formula. □

Corollary 7.8 (Rank-pattern version)**.** For $\rho \in \mathcal{R}_{d,S}^r$, define

$$m_{ab}(\rho) := \rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1},$$

and

$$\mathsf{E}(\rho) := \sum_{(a,b),(c,d)} m_{ab}(\rho) m_{cd}(\rho)\, \mathbf{1}_{[a,b]\subseteq[c,d]}.$$

Then

$$\operatorname{codim}_\Omega \Gamma_{d,S}^r[\rho] = \Big(\sum_{\ell=1}^{L} d_\ell d_{\ell-1} - \sum_{\ell=0}^{L} d_\ell^2 + (d_L - r)(d_0 + r)\Big) + \mathsf{E}(\rho),$$

and

$$\operatorname{codim}_\Omega \mathcal{C}_{d,S}^r[\rho] = \Big(\sum_{\ell=1}^{L} d_\ell d_{\ell-1} - \sum_{\ell=0}^{L} d_\ell^2 + d_0 d_L\Big) + \mathsf{E}(\rho).$$

Proof. This is the multiplicity formula of Theorem 7.7 rewritten via the inversion formula of Proposition 5.4. □

Corollary 7.9 (Whole-locus codimension formulas)**.** One has

$$\operatorname{codim}_\Omega \Gamma^r_{d,S} = \Big(\sum_{\ell=1}^{L} d_\ell d_{\ell-1} - \sum_{\ell=0}^{L} d_\ell^2 + (d_L - r)(d_0 + r)\Big) + \min_{\rho \in \mathcal{R}^r_{d,S}} \mathsf{E}(\rho),$$

and

$$\operatorname{codim}_\Omega \mathcal{C}^r_{d,S} = \Big(\sum_{\ell=1}^{L} d_\ell d_{\ell-1} - \sum_{\ell=0}^{L} d_\ell^2 + d_0 d_L\Big) + \min_{\rho \in \mathcal{R}^r_{d,S}} \mathsf{E}(\rho).$$

Equivalently,

$$\operatorname{codim}_\Omega \mathcal{C}^r_{d,S} = \operatorname{codim}_\Omega \Gamma^r_{d,S} + r(d_0 - d_L + r).$$

Proof. Each locus is a finite union of orbits, so its dimension is the maximum of the dimensions of its strata. Equivalently, its codimension is the minimum of the codimensions of its strata. Apply Corollary 7.8 and subtract the two constants. □

Corollary 7.10 (Full-rank case)**.** If $r = d_L$, then

$$\Gamma^{d_L}_{d,S} = \Sigma^{d_L}_d$$

is open in Ω, hence

$$\operatorname{codim}_\Omega \mathcal{C}^{d_L}_{d,S} = d_0 d_L.$$

Proof. The full-rank determinantal locus is open under the no-bottleneck assumption, so

$$\operatorname{codim}_\Omega \Gamma^{d_L}_{d,S} = 0.$$

Now apply Corollary 7.9. □

8 Discussion and Future Work

The critical loci of deep linear networks are not discrete, so the usual language of isolated saddles is not appropriate. Theorems 5.9 and 6.2 provide the missing discrete structure: every critical orbit of rank r carries a unique cyclic rank pattern. This is the combinatorial identity of a saddle stratum.

The direct weight-space formulation is important here. It shows that the cyclic equations, the critical condition, and the relevant orbit problems all live in the same parameter space Ω. The cyclic locus is not just an auxiliary construction; it is the ambient space whose orbit combinatorics transfers directly to the critical locus.

The codimension formulas of Section 7 suggest several further directions. First, the function $\mathsf{E}(\rho)$ gives a finite optimization problem for the codimension of the critical locus, and it is natural to ask for a more compact closed form. Second, the partial order of orbit closures in $\mathcal{R}^r_{d,S}$ should encode how saddle strata degenerate into one another. Third, the codimension shift

$$r(d_0 - d_L + r)$$

between the cyclic and critical loci is the first indication of an "RLCT-type" contribution associated with saddles. Unlike the real log canonical threshold at a minimum, no standard invariant is currently available for a saddle locus, but the present stratification suggests a natural combinatorial proxy.

Finally, the practical motivation is dynamical. In deep linear networks, the training trajectory of stochastic gradient descent interacts with critical loci rather than isolated points. A finite combinatorial model of saddle strata is a first step toward understanding how these loci shape optimization trajectories.

A Proof of the Inversion Formula

We give a slightly more explicit version of the proof of Proposition 5.4. For every admissible interval (a, b), the forward formula gives

$$\rho_{ab} = \sum_{u \le a \le b \le v} m_{uv}.$$

Therefore

$$\rho_{ab} - \rho_{a-1,b} - \rho_{a,b+1} + \rho_{a-1,b+1} = \sum_{u,v} m_{uv}\, \Xi_{uv},$$

where

$$\Xi_{uv} := \mathbf{1}_{u \le a \le b \le v} - \mathbf{1}_{u \le a-1 \le b \le v} - \mathbf{1}_{u \le a \le b+1 \le v} + \mathbf{1}_{u \le a-1 \le b+1 \le v}.$$

If $u < a$, then the first two indicators cancel, and likewise the last two indicators cancel. If $v > b$, then the first and third indicators cancel, and likewise the second and fourth indicators cancel. The only way for a nonzero coefficient to remain is therefore

$$u = a, \qquad v = b.$$

In that case

$$\Xi_{ab} = 1.$$

Thus the entire sum reduces to m_{ab}, proving the inversion formula.

Lean verification. This proposition has been checked in Lean.

B Projectives, Syzygies, and the Ext^1 Criterion

This appendix records the homological comparison with the older codimension approach.

Proposition B.1 (First syzygy of an interval module)**.** Let $M(a, b)$ be a nonprojective interval module. Define

$$\beta(a) := \begin{cases} L, & a = 0, \\ a + L - 1, & 1 \le a \le L. \end{cases}$$

Then there is a short exact sequence

$$0 \to M(b + 1, \beta(a)) \to P(a) \to M(a, b) \to 0.$$

Proof. By Corollary 4.6, the projective cover of $M(a, b)$ is

$$P(a) = M(a, \beta(a)).$$

The kernel is the tail interval remaining after quotienting by the first $b - a + 1$ composition factors, namely $M(b + 1, \beta(a))$. □

Corollary B.2 (Ext^1-criterion)**.** For admissible intervals (a, b) and (c, d),

$$\dim_{\mathbb{R}} \mathrm{Ext}^1_A(M(a, b), M(c, d)) = \begin{cases} 1, & M(a, b) \text{ nonprojective and } [b + 1, \beta(a)] \subseteq [c, d], \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Apply $\mathrm{Hom}_A(-, M(c, d))$ to the short exact sequence above and use Proposition 7.1. □

Remark B.3 (Comparison with the old codimension strategy)**.** The older residual-space approach expressed codimensions by deformation theory and Ext^1-spaces. In the present paper the orbit decomposition is already exact, so the codimension formulas of Section 7 are obtained more directly from group dimensions and endomorphism spaces. The Ext^1-criterion remains useful as a consistency check, but it is no longer needed for the main theorem.

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